

Exploratory Data Analysis Report

Titanic Dataset — Task 5, Data Analyst Internship (Elevate Labs)

Dataset	Titanic (891 passengers, 12 columns)
Tools	Python — Pandas, Matplotlib, Seaborn
Objective	Extract insights using visual & statistical exploration

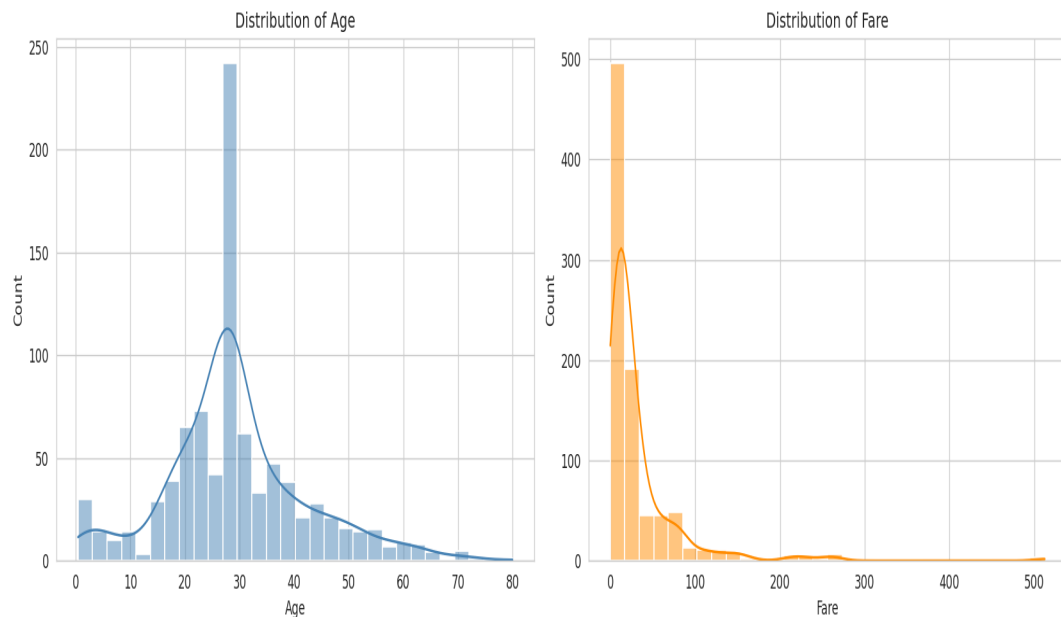
1. Dataset Overview

The Titanic dataset contains 891 passenger records across 12 columns, including demographic details (Age, Sex), travel details (Pclass, Fare, Cabin, Embarked), family details (SibSp, Parch), and the target variable Survived. Missing values were found in Age (~20%), Cabin (~77%), and Embarked (2 records). For this EDA, Age and Fare were median-imputed and Embarked was mode-imputed to allow clean visualization; Cabin was left as-is due to its high missingness.

Metric	Value
Total Passengers	891
Survived	342 (38.4%)
Did Not Survive	549 (61.6%)
Missing: Age	177 (19.9%)
Missing: Cabin	687 (77.1%)
Missing: Embarked	2 (0.2%)

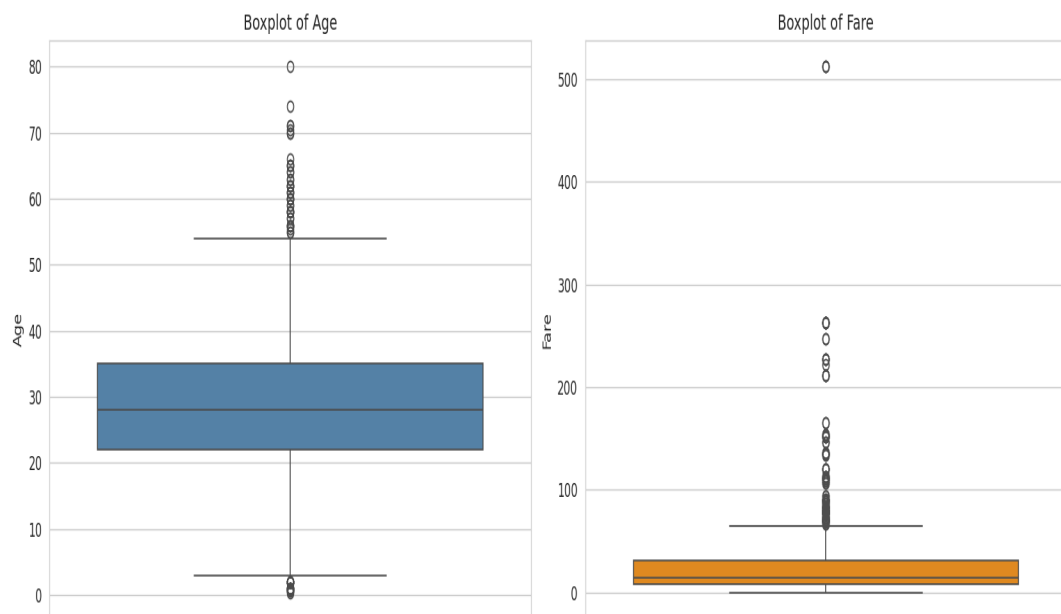
2. Univariate Analysis

2.1 Distribution of Age and Fare



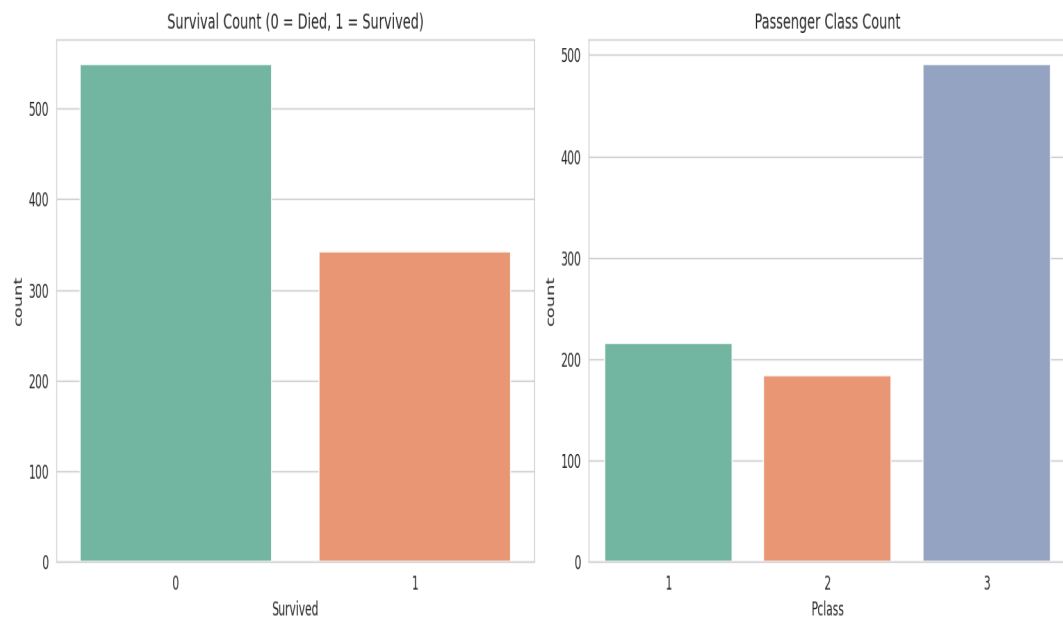
Observation: Age is roughly bell-shaped, concentrated between 20-40 years, with a slight right skew from older passengers. Fare is heavily right-skewed: most passengers paid low fares, but a small group paid very high fares, creating a long tail typical of 1st class tickets.

2.2 Boxplots of Age and Fare



Observation: The Fare boxplot shows numerous outliers above the upper whisker, confirming the right-skew from the histogram. Age shows a few high-end outliers (elderly passengers) but is otherwise well contained.

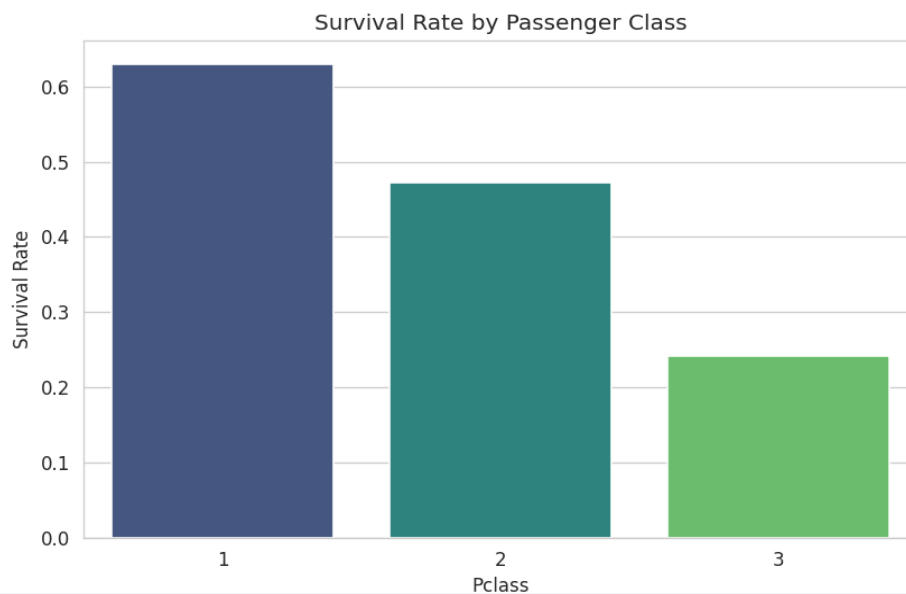
2.3 Survival and Passenger Class Counts



Observation: More passengers died (549) than survived (342), a roughly 62/38 split. Third class was the largest group of passengers, followed by first and then second class.

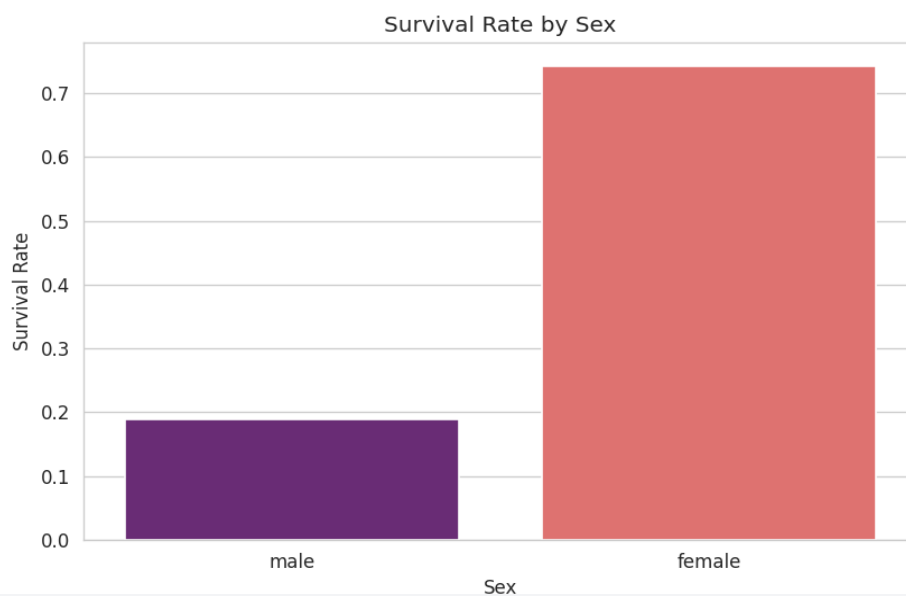
3. Bivariate Analysis

3.1 Survival Rate by Passenger Class



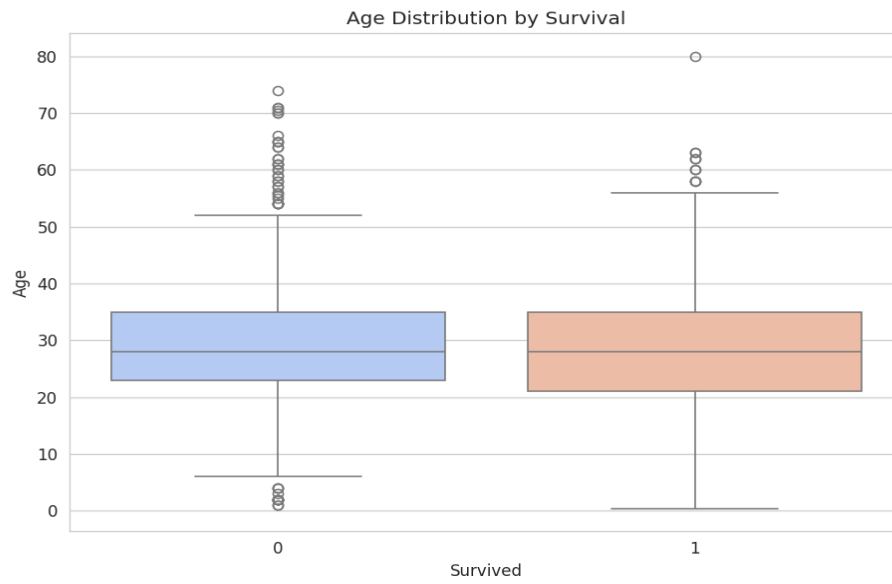
Observation: Survival rate drops sharply from 1st class (~63%) to 3rd class (~24%), suggesting socio-economic status (proxied by ticket class) played a major role in survival, likely tied to cabin location and lifeboat access priority.

3.2 Survival Rate by Sex



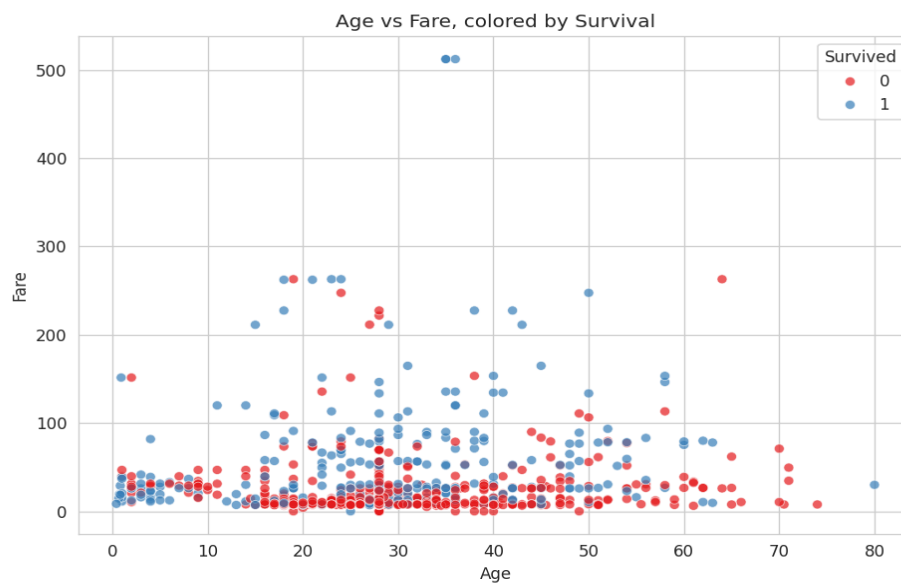
Observation: Female passengers survived at ~74% versus ~19% for males — by far the strongest single relationship in the dataset, consistent with the historical "women and children first" evacuation protocol.

3.3 Age Distribution by Survival



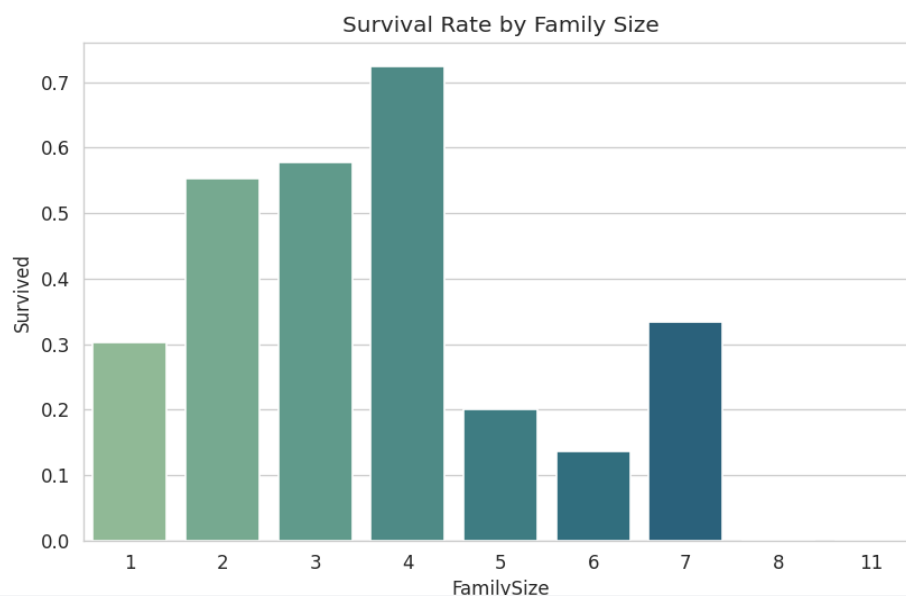
Observation: Median age is similar between survivors and non-survivors, but survivors show a slightly wider spread toward younger ages, consistent with children being prioritized during evacuation.

3.4 Age vs. Fare, Colored by Survival



Observation: There is no strong linear relationship between Age and Fare. However, most high-fare passengers (upper region of the plot) survived, reinforcing that higher fare — correlated with class — improved survival odds largely independent of age.

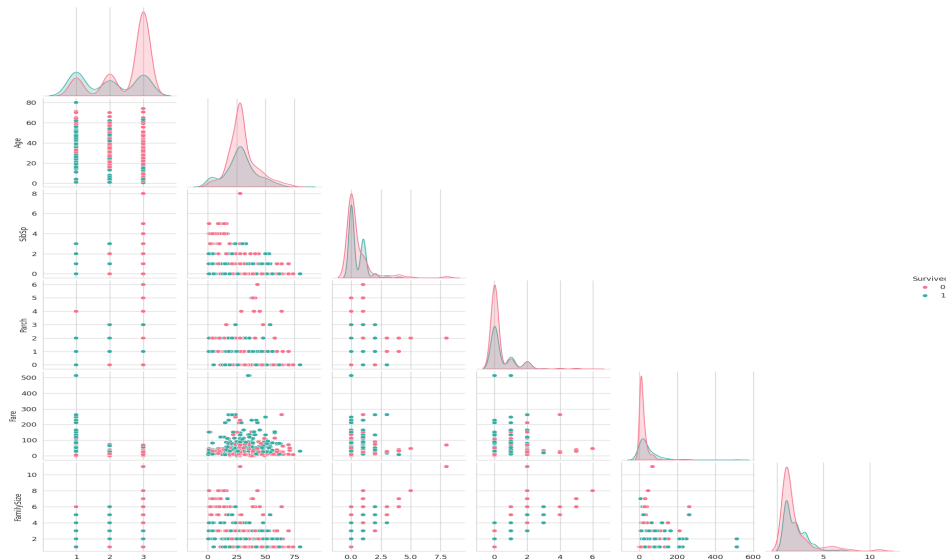
3.5 Survival Rate by Family Size



Observation: Passengers traveling completely alone or in very large families (6+) had lower survival rates, while small families of 2-4 members had the best survival odds, balancing the benefit of assistance evacuating against the difficulty of moving a large group quickly.

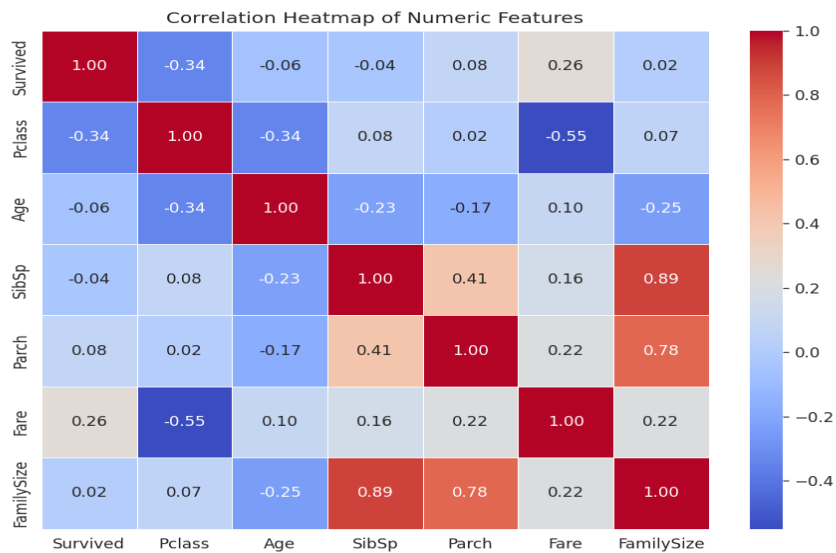
4. Multivariate Analysis

4.1 Pairplot of Key Numeric Features



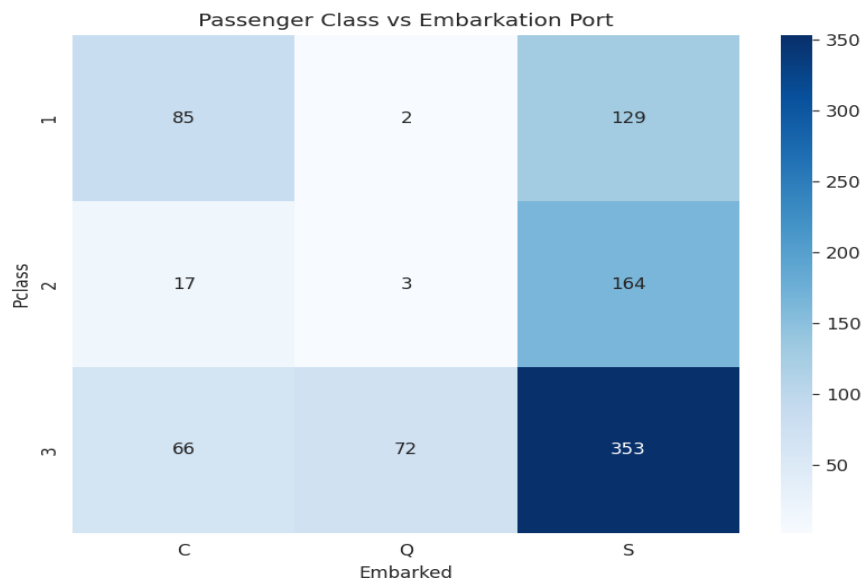
Observation: The pairplot scans all numeric variable relationships at once. Pclass and Fare show a visible inverse relationship (lower class number pays more). Diagonal density plots show survivors skew toward lower Pclass values and higher Fare, reinforcing the bivariate findings above.

4.2 Correlation Heatmap



Observation: Fare has the strongest positive correlation with Survived (~ 0.26), while Pclass has the strongest negative correlation (~ -0.34). SibSp and Parch correlate with each other and with FamilySize by construction — not a meaningful multicollinearity concern for this EDA.

4.3 Passenger Class vs. Embarkation Port



Observation: Most passengers embarked from Southampton (S) across all classes. Cherbourg (C) shows a notably higher proportion of 1st class passengers relative to the other two ports.

5. Summary of Findings

- Class matters most: survival falls from ~63% (1st class) to ~24% (3rd class); Pclass has the strongest negative correlation with survival among numeric features.
- Sex is the single biggest predictor: females survived at ~74% vs. ~19% for males.
- Fare correlates positively with survival, acting as a proxy for socio-economic status and cabin location.
- Family size has a sweet spot: small families (2-4) outperformed solo travelers and very large families.
- Age is roughly normally distributed (20-40 concentration); children show a modestly higher survival tendency.
- Fare is right-skewed with high-value outliers; a log transform is recommended before use in linear models.
- Cabin (~77% missing) is largely unusable directly and would need engineering (e.g. a "has cabin" flag); Age (~20% missing) was median-imputed for this EDA.
- No problematic multicollinearity among core numeric features beyond the expected SibSp/Parch/FamilySize relationship, which exists by construction.

Conclusion: Survival on the Titanic was driven primarily by **sex**, **passenger class**, and **fare** — all pointing to unequal evacuation access along social and gender lines — with **family size** playing a secondary role.